EXCEL + PYTHON PROJECT · RETAIL

Retail Sales Performance Analysis

6 months of sales for a 4-store chain, analysed to set next quarter's targets

Excel | Pivot Tables | Python | Pandas | KPI Analysis

Rs 52.32L
revenue analysed

31 / 72
store-category-months hit target

Rs 2.8L
opportunity found in one store

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THE BUSINESS PROBLEM

"Are our stores really on target?"

STAKEHOLDER

Head of Retail, 4 stores: Mumbai, Pune, Nashik, Nagpur

DECISION

Set **Q1 2027 targets** per store and category, and decide where the **marketing budget** goes

PROBLEM STATEMENT

We don't know which stores and categories miss target, or whether the cause is fewer visitors, lower conversion or smaller bills.

APPROACH: 7-STEP ANALYST WORKFLOW

1. Define	problem statement, clarifying questions
2. Collect	data dictionary, grain
3. Clean	Python / Pandas
4. Explore	store & month vs target
5. Analyze	Excel KPIs, driver breakdown
6. Visualize	Excel dashboard
7. Communicate	insights, actions, exec summary

STEP 2 · COLLECT

72 rows: 4 stores × 3 categories × 6 months

COLUMN	MEANING	TYPE
Month	Sales month, May to Oct 2026	Date
Store / Region	Store city and zone	Nominal
Category	Apparel, Footwear, Accessories	Nominal
Visitors	People entering the store	Discrete
Orders	Bills / purchases	Discrete
Revenue / Target	Monthly sales and target in Rs	Continuous

KPIs I CALCULATED

Conversion % = Orders ÷ Visitors

AOV = Revenue ÷ Orders

Variance % = (Revenue − Target) ÷ Target

Driver tree: Revenue = Visitors × Conversion × AOV. This tells us *why* a store misses, not just *that* it misses.

STEP 3 · CLEANING

From 74 messy rows to 72 trusted rows

PYTHON (PANDAS)

```
df = df.drop_duplicates()
df['Store'] = df['Store'].str.strip()

# grain check: each store needs 18 rows
df.groupby('Store').size()
# Mumbai 16, Pune 19, Nagpur 19 -> 2 rows mislabelled

df['Revenue'] = pd.to_numeric(
    df['Revenue'].str.replace('Rs','')
                 .str.replace(',',''))

med = df.groupby(['Store','Category'])
        ['Visitors'].transform('median')
df['Visitors'] = df['Visitors'].fillna(med)

df['Month'] = pd.to_datetime(df['Month'],
                             format='%b-%Y')
```

ISSUES FIXED

2 duplicate rows	revenue counted twice
2 rows tagged to wrong store	Mumbai short by ~Rs 1.4L
1 wrong category label	wrong category totals
Revenue as text "Rs 98,927"	can't be summed
2 missing visitor counts	wrong conversion
Month as text	wrong sort order

Lesson: fixing spellings wasn't enough. A **row-count check per store** exposed 2 rows under the wrong store, found using Region and Target as evidence.

STEP 4 · EXPLORE

On target overall, but it hides the misses

Rs 52.32L
revenue vs Rs 52.32L target

0.0%
overall variance

2.68%
conversion rate

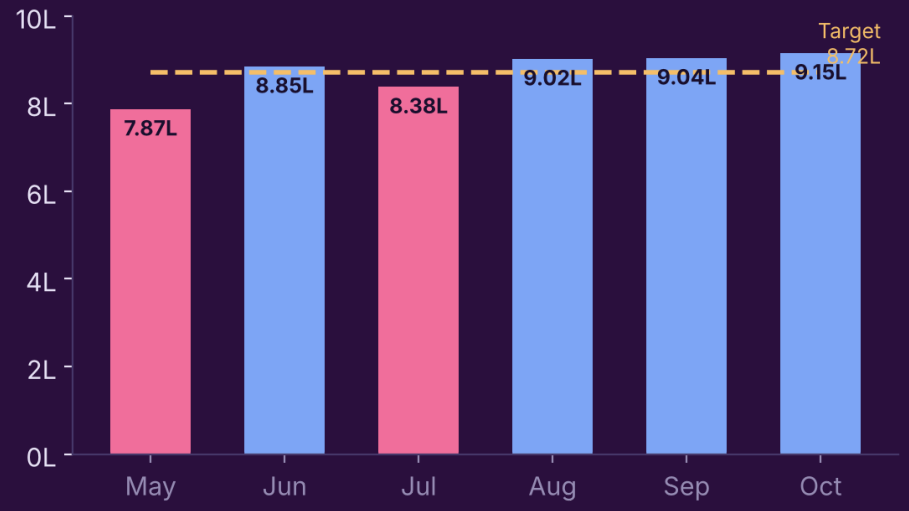
31 / 72
store-category-months hit target (43%)

WHAT THE DATA SHOWS

- **Nagpur** was the weakest store: Rs 14.96L vs Rs 15.12L target **(−1.1%)**. Mumbai also missed **(−0.6%)**.
- **Nashik** **(+2.1%)** and **Pune** **(+0.7%)** beat target.
- **May** was the weakest month: Rs 7.87L vs Rs 8.72L **(−9.7%)**. July also missed (−3.9%).
- More than half of all store-category-months missed target, so one flat company target is not enough.

STEP 5 · WHEN DID WE MISS?

May was a footfall problem, not a sales problem



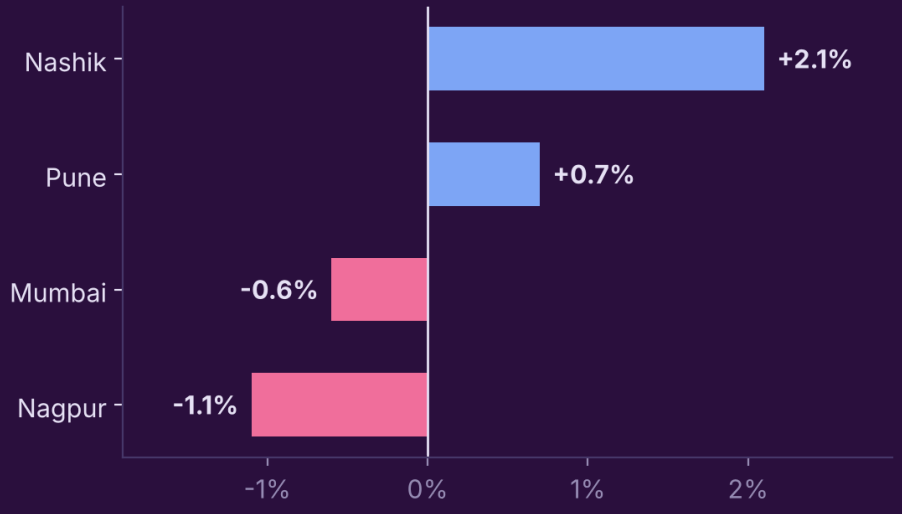
−9.7%
May vs target (Rs 84,635 short)

33,273
May visitors, lowest month (Aug to Oct avg ~38,800)

All 4 stores missed in May, but conversion was normal (2.65%). **Fewer people came in**, so the fix is footfall, not the store team.

STEP 5 · WHO MISSED?

Two stores below target, two above



REVENUE VS TARGET, MAY TO OCT

Nashik	Rs 7.91L	+2.1%
Pune	Rs 12.33L	+0.7%
Mumbai	Rs 17.12L	−0.6%
Nagpur	Rs 14.96L	−1.1%

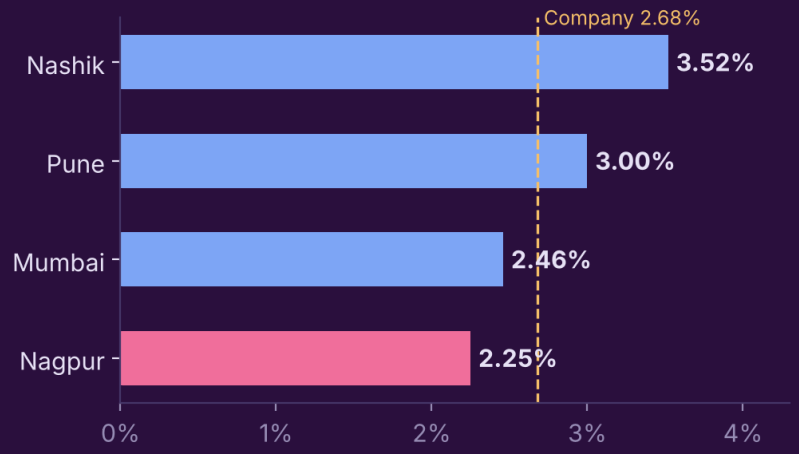
Gaps look small in %, so I charted **variance %** instead of raw revenue. On a Rs 17 lakh scale the bars look identical and hide the story.

STEP 5 · WHY DOES NAGPUR MISS?

Nagpur gets the customers but doesn't convert them

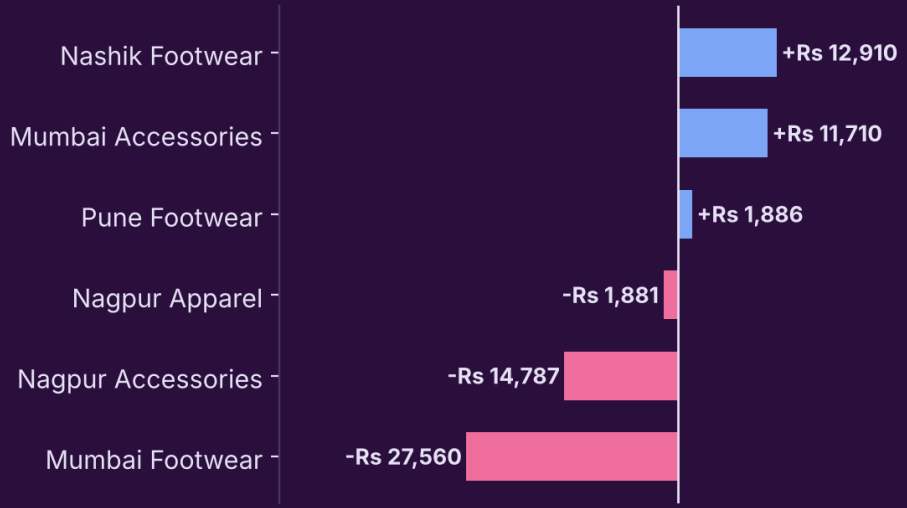
STORE	VISITORS	CONVERSION	AOV
Mumbai	77,208	2.46%	Rs 901
Nagpur	59,544 #2	**2.25% · lowest**	**Rs 1,115 · highest**
Nashik	31,785	**3.52%**	Rs 707
Pune	50,998	3.00%	Rs 805

At the company conversion rate (2.68%), Nagpur would get ~255 more orders × Rs 1,115 = **~Rs 2.8 lakh more revenue**, far more than its Rs 16,303 gap to target.



STEP 5 · WHERE EXACTLY?

Two store-category combos explain most of the miss



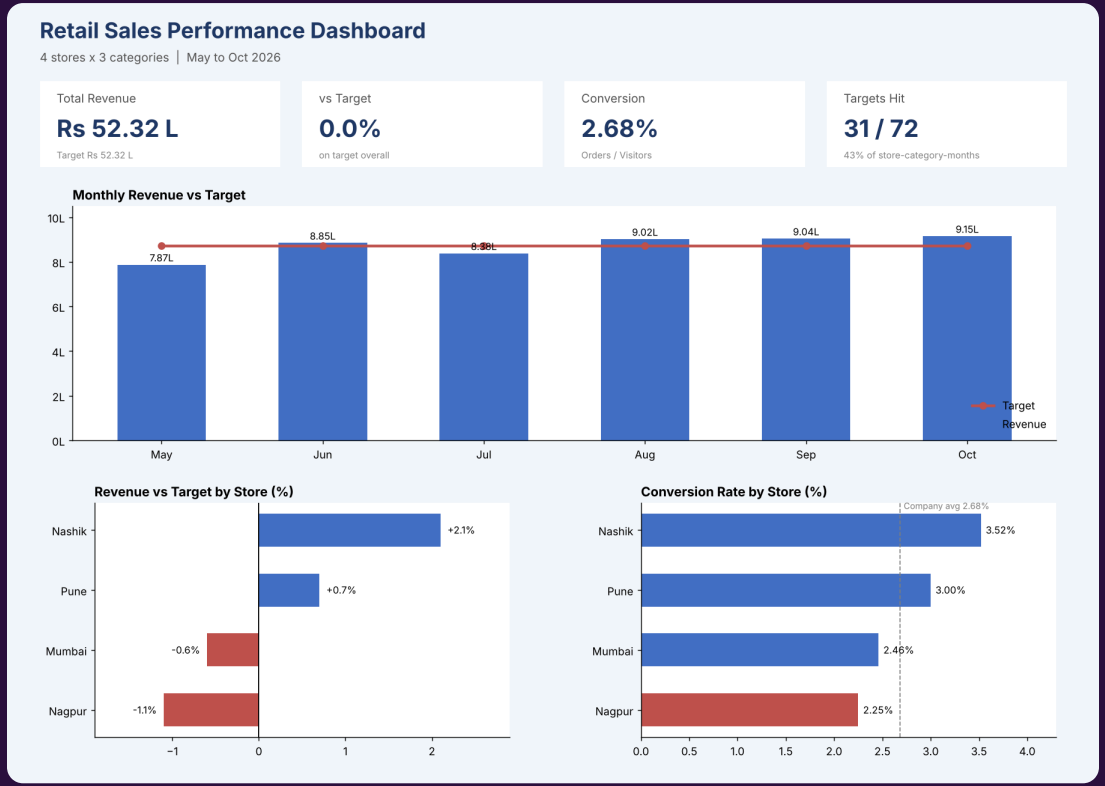
−Rs 27,560
Mumbai Footwear, the largest single gap

−8.2%
Nagpur Accessories vs target

Meanwhile **Nashik Footwear beat target by Rs 12,910**. Same category, different store, so the playbook already exists inside the company.

STEP 6 · VISUALIZE

One-page Excel dashboard: when, who, why



BUILT IN EXCEL

- Calculated KPI columns
- Pivot tables by store and month
- KPI cards
- Combo chart: revenue vs target line
- Variance and conversion bar charts

Ratios were **recalculated from totals**, never summed in the pivot. Summing 18 monthly rates gave a fake "44% conversion".

STEP 7 · RECOMMENDATIONS

What to do in Q1 2027

INSIGHT	IMPACT	ACTION
Nagpur converts 2.25% vs 2.68% company	**~Rs 2.8L upside**	Store manager: check stock-outs and floor staff at peak hours, review accessories pricing. Q1 target 2.6%
May −9.7%, all stores missed, visitors lowest	**Rs 84,635 in one month**	Marketing: April footfall campaign before May. Target 37,000+ May visitors
Mumbai Footwear −Rs 27,560 while Nashik Footwear +Rs 12,910	**~Rs 42,000 from 2 combos**	Mumbai manager: copy Nashik's footwear display and stock mix. Set Q1 targets per store-category

Check next: Nagpur stock-out and staffing data, May marketing activity, and last year's data to confirm seasonality.

EXECUTIVE SUMMARY

Situation · Complication · Resolution

SITUATION

4 stores earned **Rs 52.32 lakh** in May to Oct 2026, exactly on target.

COMPLICATION

Only **31 of 72** store-category-months hit target. **May (−9.7%)** and **Nagpur (−1.1%)** were the weak spots, and Nagpur's problem is **low conversion**, not footfall.

RESOLUTION

(1) April footfall campaign · (2) Mumbai copies Nashik's footwear playbook · (3) Fix Nagpur's conversion, worth ~Rs 2.8 lakh

7-step workflow | KPI driver tree | Variance analysis | SCR storytelling